

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS: Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

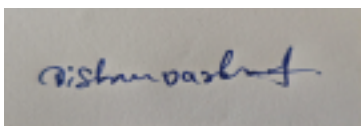
Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the

Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I (B.VishnuVardhan / 192311267) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete	Good understanding with minor gaps	Basic understanding with some	Poor or incorrect understanding

		understanding of the concept		misconceptions	
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor	Covers only some parts	Incomplete or missing response

			omissions		
Clarity& Presentation	10	Clear, concise, and well-organiz ed answer	Understan da ble with minor issues	Somewhat unclear or poorly structured	Difficult to understa nd

Responsive web design

Responsive web design (RWD) is a web design approach development approach that enable website automatic adjust layout images and content the main objective of response it's design to provide user attractive and user-friendly experience across all devices without creating separate website.

Importance of (RWD) :

- Provides a better user experience on all screen sizes.
- Ensures compatibility with mobile, tablet, and desktop devices.
- Reduces website maintenance by using a single responsive design.
- Increases website accessibility, reliability and consistency.

CSS techniques

- media queries : Apply different CSS rules based on screen width or device characteristics.
- flexible layouts : use relative units such as % , em, rem and vw instead of fixed pixel values.
- Responsive image : Set image using max-width: 100%; height: auto; so they scale properly.
- viewport meta Tag : Ensures the webpage scales correctly on mobile devices.

Example 1

HTML

```
<div class="container">
```

```
<div class="box">Home</div>
```

```
<div class="box">About</div>
```

```
<div class="box">Contact</div>
```

```
</div>
```

improvements

- use larger buttons and touch-friendly navigation for mobile.
- optimize images and files to improve loading speed.
- maintain readable font sizes and sufficient spacing.
- Test the website on different devices and browser to ensure consistent performance.

Conclusion :- Responsive web design is essential for modern websites because ensures that content is accessible and visually appealing on mobile, tablet, and desktop devices. By using CSS techniques such as media queries, flexible units, responsive image, and Flexbox or Grid layouts, developers can build adaptive website that improve usability, performance, and overall user experience.

② client side validation:

- client-side validation checks user input in the browser to catch errors early.
- like empty field, wrong email format or invalid phone numbers.
- it improved user experience and reduced unnecessary server load but should not replace server-side validation.

JavaScript event handling:-

Events like submit blur or input trigger validation function that check field values and show error messages instantly.

Regular Expression; (Reg ex)

used to match pattern -eg. checking if an email has text@text.com

Format or a phone number only digits of correct length.

Example 2

HTML

<form id="my form">

<input type="text" id="email" placeholder="enter email">

<button type="submit">Submit</button>

</form>

Script

```
if (!pattern.test(email)) {
```

```
    error.textContent = "(Invalid email format)";
```

```
} else {
```

```
    alert("Form Submitted Successfully!");
```

```
}
```

```
</script>
```

Important:

Add real time validation on input/blur event.

Disable submit button until valid algory, i.e.

validity on server too and are marked

attributes for accessibility.

3) CSS Box Model, Positioning media Queries.

Box model:-

- Content → actual text/image
- padding → space between content and border.
- Border → edge around padding.
- margin → space outside the border.

CSS

```
div {
```

```
width: 200px;
```

```
padding: 10px;
```

```
border: 5px solid black
```

```
margin: 15px;
```

```
box-sizing: border-box;
```

```
}
```

CSS Positioning:-

STATIC default relative, absolute, fixed,

sticky wrong use of absolute / fixed often caused layout breaking on resize.

Causes of layout Breaking:-

Fixed pixel width, missing box-sizing: border-box, overflow content.

Flx with media queries:

- Containers display: flex;

@media (max-width: 768px) {

• container & flex-direction: column;

}

Best Practices:

use relative units,

→ box-sizing: border-box

globally, Flexbox Grid instead of absolute positioning, test at multiple break points.

avoid fixed width on containers.

4) DOM manipulation for dynamic content:

→ The DOM (Document Object Model) is a tree-like representation of an HTML page that JavaScript can access and modify. It allows dynamic updates to content, structure, and style.

Without reloading the page

JavaScript DOM manipulation:

Methods like

- getElementById
- querySelector
- innerHTML
- createElement
- appendChild
- addEventListener

Let's see how to update the DOM in real-time (e.g. updating a counter, adding list items, changing text on button clicks).

